

练习题 12.1

12.1.1 求下列极限:

(1) $I = \lim_{x \rightarrow 0} \int_{-1}^1 \sqrt{x^2 + y^2} dy$; (2) $I = \lim_{\beta \rightarrow 0} \int_0^2 x^2 \cos \beta x dx$;

(3) $I = \lim_{\alpha \rightarrow 0} \int_{\alpha}^{\alpha + \frac{\pi}{2}} \frac{\sin^2 x}{4 + \alpha x^2} dx$.

(1) $\int_{-1}^1 |y| dy = 1$ (2) $\int_0^2 x^2 dx = \frac{8}{3}$ (3) $\frac{1}{4} \int_0^{\frac{\pi}{2}} \sin^2 x dx = \frac{\pi}{16}$

12.1.2 求下列函数的导数:

(1) 设 $\varphi(x) = \int_{a+x}^{b+x} \frac{\sin xy}{y} dy$, 求 $\varphi'(x)$;

(2) $F(\alpha) = \int_0^{\alpha^2} \left(\int_{x-\alpha}^{x+\alpha} \sin(x^2 + y^2 - \alpha^2) dy \right) dx$, 求 $F'(\alpha)$;

(3) $F(x) = \int_a^b f(y) |x - y| dy$, 其中 $a < b$, $f(y)$ 为 $[a, b]$ 上可微函数, 求 $F''(x)$.

(1) $\varphi'(x) = \int_{a+x}^{b+x} \cos xy \cdot dy + \frac{\sin(b+x)x}{x} - \frac{\sin(a+x)x}{x}$
 $= \frac{\sin xy}{x} \Big|_{a+x}^{b+x} + \dots$

(2) $\frac{-\cos(x^2 + y^2 - \alpha^2)}{2y} \Big|_{x-\alpha}^{x+\alpha} = \int_0^{\alpha^2} \left[\frac{\cos(2x^2 - 2\alpha)}{2(x-\alpha)} - \frac{\cos(2x^2 + 2\alpha)}{2(x+\alpha)} \right] dx$

$\int_{x-\alpha}^{x+\alpha} \sin(x^2 + y^2 - \alpha^2) dy = g(x)$

$F(\alpha) = \int_0^{\alpha^2} g(x, y, \alpha) dx$

$F'(\alpha) = 2\alpha \cdot g(\alpha^2, y, \alpha) + \int_0^{\alpha^2} \frac{\partial g}{\partial \alpha} dx$

$\frac{\partial g}{\partial \alpha} = -2\alpha \int_{x-\alpha}^{x+\alpha} \cos(x^2 + y^2 - \alpha^2) dy + \sin(2x^2 + 2\alpha) + \sin(2x^2 - 2\alpha)$

$2\sin 2x^2 \cdot 2\alpha x$
 $2\cos 2x^2 \sin 2\alpha x$

$F'(\alpha) = 2\alpha \cdot \int_{\alpha^2-\alpha}^{\alpha^2+\alpha} (\alpha^4 - \alpha^2 + y^2) dy - 2\alpha \cdot \int_0^{\alpha^2} dx \int_{x-\alpha}^{x+\alpha} \cos(x^2 + y^2 - \alpha^2) dy + 2 \int_0^{\alpha^2} \cos 2x^2 \sin 2\alpha x dx$

(3) $F(x) = \int_a^b f(y) |x-y| dy$, 其中 $a < b$, $f(y)$ 为 $[a, b]$ 上可微函数, 求 $F''(x)$.

12.1.3 利用积分号下求导数求下列积分:

$$x-y=t, \quad y=x-t$$

$$- \int_{x-a}^{x-b} f(x-t) \cdot |t| \cdot dt$$

$$\therefore F'(x) = - \int_{x-a}^{x-b} f'(x-t) \cdot |t| dt - f(b) \cdot |x-b| + f(a) \cdot |x-a|$$

① $x > b, |x| < a$

$$\begin{aligned} F''(x) &= - \int_{x-a}^{x-b} f''(x-t) \cdot |t| dt - f'(b) |x-b| + f'(a) |x-a| - f(b) + f(a) \\ &= \int_{x-a}^{x-b} |t| df'(x-t) \dots \end{aligned}$$

$$= |t| \cdot f'(x-t) \Big|_{x-a}^{x-b} - \int_{x-a}^{x-b} f'(x-t) dt \dots$$

$$= 0$$

$$\textcircled{2} x \in (a, b) \quad F'(x) = - \int_{x-a}^{x-b} f'(x-t) |t| dt - f(b)(b-x) + f(a)(x-a)$$

$$F''(x) = - \int_{x-a}^{x-b} f''(x-t) \cdot |t| dt - f'(b)(b-x) + f'(a)(x-a) + f(b) + f(a)$$

$$= f'(x-t) |t| \Big|_{x-a}^{x-b} - \int_{x-a}^{x-b} f'(x-t) dt \dots$$

$$= f(b) - f(a) + f(b) + f(a)$$

$$= 2f(b)$$

?

12.1.3 利用积分号下求导数求下列积分:

$$(1) I(a) = \int_0^{\frac{\pi}{2}} \frac{\arctan(a \tan x)}{\tan x} dx;$$

$$(2) I(\theta) = \int_0^{\pi} \ln(1 + \theta \cos x) dx \quad (|\theta| < 1);$$

$$(3) I = \int_0^{\frac{\pi}{2}} \ln \frac{a + b \sin x}{a - b \sin x} \cdot \frac{dx}{\sin x} \quad (a > b > 0).$$

(1) 注意到 $f(a, x) = \frac{\arctan(a \tan x)}{\tan x}$ 与 $\frac{\partial f}{\partial a}$ 均在 $(0, \frac{\pi}{2})$ 上连续

$$I(a) = \int_0^{\frac{\pi}{2}} \frac{\arctan(a \tan x)}{\tan x} dx$$

$$\begin{aligned} I'(a) &= \int_0^{\frac{\pi}{2}} \frac{1}{1+a^2 \tan^2 x} dx = \int_0^{\frac{\pi}{2}} \frac{1}{1+a^2 \tan^2 x} \cdot \frac{1}{\tan^2 x + 1} \cdot \sec^2 x dx \\ &= \int_0^{+\infty} \frac{1}{1+a^2 t^2} \cdot \frac{1}{t^2+1} dt \end{aligned}$$

$$\frac{a^2}{1+a^2 t^2} - \frac{1}{t^2+1} = \frac{a^2(t^2+1) - a^2 t^2}{(1+a^2 t^2)(t^2+1)} = \frac{a^2-1}{(1+a^2 t^2)(t^2+1)}$$

$$I'(a) = \frac{1}{a^2-1} \cdot \int_0^{+\infty} \left(\frac{a^2}{1+a^2 t^2} - \frac{1}{t^2+1} \right) dt$$

$$= \frac{1}{a^2-1} \cdot (a \arctan at - \arctan t) \Big|_0^{+\infty}$$

$$= \frac{\pi}{2} \frac{1}{a+1}$$

$$\therefore I(a) = \frac{\pi}{2} \ln |a+1| + C$$

注意到 $I(0)=0$, $\therefore I(a) = \frac{\pi}{2} \ln |a+1|$

(注意 $a < 0$ 的情况)

$I(a)$ 为奇函数)

$$(2) I(\theta) = \int_0^\pi \ln(1 + \theta \cos x) dx \quad (|\theta| < 1);$$

$$(3) I = \int_0^{\frac{\pi}{2}} \ln \frac{a + b \sin x}{a - b \sin x} \cdot \frac{dx}{\sin x} \quad (a > b > 0).$$

12.1.4 利用对参数的积分法求下列积分:

$$\begin{aligned} I'(\theta) &= \int_0^{2\pi} \frac{\cos x}{1 + \theta \cos x} dx = \int_0^{2\pi} \frac{1}{1 + \theta \cos x} d \sin x = \int_0^{2\pi} \sin x \cdot \frac{-\theta \sin x}{(1 + \theta \cos x)^2} dx = \int_0^{2\pi} \frac{-\theta \sin^2 x}{(1 + \theta \cos x)^2} dx \\ &= - \int_0^{2\pi} \frac{\theta}{(1 + \theta \cos x)^2} dx + \int_0^{2\pi} \frac{\theta \cos^2 x}{(1 + \theta \cos x)^2} dx \\ \int_0^{2\pi} \frac{\cos x}{1 + \theta \cos x} dx &= \frac{1}{\theta} \cdot \int_0^{2\pi} \left(1 - \frac{1}{1 + \theta \cos x} \right) dx \\ &= \frac{2\pi}{\theta} - \frac{2\pi}{\sqrt{1 - \theta^2}} \end{aligned}$$

$$\therefore I(\theta) = 2\pi \ln \theta - 2\pi \arcsin \theta$$

$$\ln |\sqrt{1 - \theta^2} + 1|$$

$$(3) I(b) = \int_0^{\frac{\pi}{2}} \ln \frac{a + b \sin x}{a - b \sin x} \cdot \frac{dx}{\sin x}$$

$$I'(b) = \int_0^{\frac{\pi}{2}} \frac{1}{\sin x} \cdot \left(\frac{\sin x}{a + b \sin x} + \frac{\sin x}{a - b \sin x} \right) dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{2a}{a^2 - b^2 \sin^2 x} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{2a(\tan^2 x + 1)}{a^2(\tan^2 x + 1) - b^2 \tan^2 x} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{2a}{a^2 + (a^2 - b^2)\tan^2 x} d\tan x$$

$$= \int_0^{+\infty} \frac{2a}{a^2 + (a^2 - b^2)x^2} dx$$

$$= \frac{2a}{a^2 - b^2} \cdot \frac{\sqrt{a^2 - b^2}}{a} \cdot \arctan \frac{\sqrt{a^2 - b^2}}{a} x \Big|_0^{+\infty}$$

$$= \frac{\pi}{\sqrt{a^2 - b^2}}$$

$$\therefore I(b) = \pi \arcsin \frac{b}{a}$$

12.2.1 证明下列积分在指定区间上是一致收敛的：

(1) $\int_0^{+\infty} e^{-t^2 x^2} \cos x dx \quad (0 < t_0 < t < +\infty);$

(2) $\int_1^{+\infty} \frac{y^2 - x^2}{(x^2 + y^2)^2} dx \quad (-\infty < y < +\infty);$

(3) $\int_1^{+\infty} t e^{-tx^2} dx \quad (0 \leq t < +\infty).$

12.2.2 证明下列积分在指定区间上是一致收敛的：

(1) $\int_1^{+\infty} e^{-tx} \frac{\sin x}{x} dx \quad (0 \leq t < +\infty); \quad (2) \int_1^{+\infty} e^{-t} \frac{\sin at}{t} dt \quad (0 \leq a < +\infty);$

(3) $\int_1^{+\infty} \frac{\cos(x^2)}{1+x^p} dx \quad (0 \leq p < +\infty); \quad (4) \int_1^{+\infty} \frac{\cos tx}{1+x} dx \quad (0 < t_0 \leq t < +\infty).$

(1) $|e^{-tx^2} \cos x| \leq e^{-tx^2} \leq e^{-t_0 x^2}$

(2) $\left| \frac{y^2 - x^2}{(x^2 + y^2)^2} \right| \leq \frac{x^2}{x^4} = \frac{1}{x^2}$

(3) $0 \leq t e^{-tx^2} = \frac{1}{x^2} t x^2 e^{-tx^2}, \lim_{x \rightarrow \infty} \frac{tx^2}{e^{tx^2}} = 0. \quad \frac{1}{x^2} t x^2 e^{-tx^2} \leq \frac{1}{x^2}, \text{收敛}$

(1) ① $t=0$ 时. 原式: $\int_1^{+\infty} \frac{\sin x}{x} dx$, 而 $|\sin x| \leq 1$, 一致有界
 $\frac{1}{x} \downarrow$ 且 $\rightarrow 0$, 故此时收敛

$$\int_1^{+\infty} \sin at dt = -\frac{1}{a} \cos at \Big|_1^{+\infty} = \left| \frac{1}{a} \cos a \cdot A - \frac{1}{a} \cos a \right|$$

$$\forall A > 0, \int_1^A \cos x^2 dx = x \cos x^2 \Big|_1^A - \int_1^A x d(-\sin x^2)$$

$$\int_1^A \cos x^2 dx = A \cos A^2 - \cos 1 + \int_1^A 2x^2 \sin x^2 dx$$

$$\leq A \cos A^2 - \cos 1 + \int_1^A 2x^2 dx$$

$$= A \cos A^2 - \cos 1 + \frac{2}{3} A^3 - \frac{2}{3}$$

$$\int_1^A \cos tx dx = \frac{\sin tx}{t} \Big|_1^A = \left| \frac{\sin At}{t} - \frac{\sin t}{t} \right| \leq \frac{2}{t_0}, \text{收敛}$$

练习题 12.3

12.3.1 利用对参数的微分法求下列积分:

(1) $I = \int_0^{+\infty} \frac{\arctan bx - \arctan ax}{x} dx \quad (b > a > 0);$

(2) $I(a) = \int_0^{+\infty} \frac{\arctan ax}{x(1+x^2)} dx \quad (a \geq 0).$

$$a \int_0^{+\infty} dx \int_a^b \frac{1}{1+x^2 y^2} dy$$

$$\forall y \in [a, b], \frac{1}{1+x^2 y^2} \leq \frac{1}{a^2 x^2}, \text{故对 } x \text{ 一致收敛}$$

$$\therefore I(y) = \int_a^b dy \int_0^{+\infty} \frac{1}{1+x^2 y^2} dx = \int_a^b \frac{2}{y} \cdot \frac{1}{y} dy = \frac{2}{y} \ln \frac{b}{a}$$

$$\frac{1}{y^2} y \cdot \arctan xy$$

$$\frac{1}{y} \arctan xy \Big|_0^{+\infty}$$

(2) 显然 $\forall \alpha \geq 0$, $I(\alpha)$ 逐点收敛

$$x \cdot \frac{1}{1+\alpha^2 x^2} \cdot \frac{1}{x(1+x^2)} = \frac{1}{(1+\alpha^2 x^2)(1+x^2)} < \frac{1}{x^2}, \text{一致收敛且连续}$$

故可以逐项求导

$$\begin{aligned} I'(\alpha) &= \int_0^{+\infty} \frac{1}{(1+\alpha^2 x^2)(1+x^2)} dx = \frac{1}{1-\alpha^2} \int_0^{+\infty} \left(\frac{1}{1+x^2} - \frac{\alpha^2}{1+\alpha^2 x^2} \right) dx = \frac{1}{1-\alpha^2} \cdot (\arctan x - \alpha \arctan \alpha x) \Big|_0^{+\infty} \\ &= \frac{1}{1-\alpha^2} \cdot \frac{\pi}{2} (1-\alpha) \\ &= \frac{1}{1+\alpha} \cdot \frac{\pi}{2} \end{aligned}$$

$$\therefore I(\alpha) = \frac{\pi}{2} \ln |\alpha+1|$$

12.3.2 利用对参数的积分法求下列积分:

$$(1) I = \int_0^{+\infty} \frac{e^{-ax^2} - e^{-bx^2}}{x^2} dx \quad (a > 0, b > 0); \quad (2) I = \int_0^{+\infty} e^{-t} \frac{\sin xt}{t} dt.$$

12.3.3 求下列积分:

$$(1) I = \int_0^{+\infty} \left(\frac{e^{-ax} - e^{-\beta x}}{x} \right)^2 dx \quad (a > 0, \beta > 0);$$

$$(2) I = \int_0^{+\infty} \frac{\ln(\alpha^2 + x^2)}{\beta^2 + x^2} dx \quad (\alpha > 0, \beta > 0).$$

12.3.4 证明函数 $F(x) = \int_0^{+\infty} e^{-(y-x)^2} dy$ 对 $x \in (-\infty, +\infty)$ 可微.

$$\int_0^t \cos yt \cdot dy$$